Fergus Hudson

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Career Objective

Final year mechatronics engineering student seeking to gain knowledge and experience in engineering, specifically within the manufacturing field. I will be able to apply the engineering skills obtained through my experience at university, an industry internship, and robotics as well as transferrable skills from my work experience effectively to solve problems.

Education

Bachelor of Engineering (Honours)
Diploma in professional Engineering Practice
University of Technology Sydney (UTS)
Major in Mechatronics Mar 2020 – Jul 2026

- **GPA 6.18/7 WAM 78.18**
- 2024 UTS Tech Festival **Best Project Overall:** Warman Robot
- 2025 UTS Tech Festival **Best Project Overall and People's Choice**: Chess Playing Cobot
- 2025 Dean's List

NSW Higher School Certificate
Barker College 2019

- ATAR of 90.3 Distinguished Achiever,
- Distinguished Achiever Mathematics Advanced
- Distinguished Achiever Design and Technology (HSC Mark: 94)
- **Semi-finalist** in the national 2020 BHP Foundation Science and Engineering Awards

Employment

University of Technology Sydney
Mechanical Design Fundamentals Studio 1 – Lead Tutor July 2025 – Present

- **Utilising communication and presentation skills to tutor labs of 35+ students** on engineering topics such as mechatronics, electronics, and mechanical design
- Assisting students designing for and operating a range of manufacturing technologies including **laser cutting and 3D printing**
- Helping students bridge theory and practice through hands-on guidance in design, fabrication, testing, and refinement
- Managing and teaching other tutors to effectively deliver content and provide student support

Aryzta Australia
Continuous Improvement and Process Engineering Intern Jun 2023 – Mar 2024

- **Full time paid position** tasked with observing current production of baked items at a large-scale bakery manufacturer, collecting and analysing data, and talking to operators to understand areas of improvement with respect to product quality, waste, and operator health and safety
- Actioned above areas of improvement where possible, through the engineering design process, to improve production efficiency
- Designed and created drawings to an AS1100 standard to be manufactured externally meeting stringent design requirements for food safety. Using methods including bent sheet metal, plate, weldments, and tube
- Tested and commissioned new parts and machines on the production floor, reiterating where required
- Employment was **extended to 9 months due to strong performance**

Robotics Mentor
Barker College Middle School VEX Oct 2022 – Jun 2023

- Teach students engineering concepts including the design process, prototyping, and problem solving
- Manage student to student interactions and communicate effectively with parents to resolve issues
- Mentored a team that successfully qualified to compete in the national VEX competition

Engineering Experience

UTS Rover Team Lead Mar 2025 – Mar 2026

- Led the UTS Rover Team, setting strategic direction and coordinating student efforts across rover design, integration, competition preparation, business development, sponsorship outreach, and procurement
- Managed operations, priorities, and collaboration within a **30+ member engineering team**
- Drove design-for-manufacture decisions by Working with the UTS Workshop to ensure manufacturability for our designs across a wide range of manufacturing techniques including**, waterjet cut and bent sheet metal, 4 axis CNC, manual lathes, and tab and slot welded assemblies**
- During tenure, the team won the **Stellar Improvement Award** and qualified for the **international US Lunabotics Competition** held at NASA
- Organised all student travel, including liaising with UTS Faculty to secure support, funding and approvals for competition participation
- Oversaw project delivery using a range of project management tools to track deadlines, manage deliverables and maintain team progress across 6 sub-teams

Warman Challenge Jul 2024 – Oct 2024

- Dedicated roughly 30 hours a week during full time university load in the completion of our team's submission to the international finals
- Designed and manufactured the chassis and drive system, a one-metre-long unfolding custom worm drive powered single DoF arm, pickup mechanism and electronics
- Heavily involved in the development, testing and iteration of the drivebase DC motor control system, utilising feedforward and cascaded PID loops to ensure straight and accurate movement
- Won the **Judges Award** for innovative design and **placed 5th** at the Warman International Finals

Technical Skills

- Extensive experience in SOLIDWORKS in a range of settings (AS1100 compliant Engineering Drawings, fully parametric design with focus on design intent and design for manufacture)
- Engineering report writing
- Problem scoping, prototyping and iterative design
- Design evaluation, validation, and inspection
- Project planning
- MATLAB, embedded systems (ESP32, STM32 Arduino), C++
- CAM experience (Fusion 360)
- Machine tools: Lathes, **CNC routers, Laser Cutters, 3D Printers,** Table Saws, hand tools
- Basic experience welding: Aluminium TIG (Tungsten Inert Gas), and MIG (Metal Inert Gas)

Transferrable Skills

Problem Solving

As UTS Rover Team Lead, and Process Engineering intern, I have worked to implement solutions to a wide range of unforeseen problems both within the time critical environment of competition and the workplace. I have passed on this experience through my experience as a tutor and robotics mentor.

Communication

Through my experience working as a robotics mentor and tutor I have acquired valuable skills in communicating confidently and efficiently with peers, clients and students, in written, verbal, and public speaking contexts.

Teamwork

Proven ability to work within a team environment, understand my peers' perspectives and work towards a common goal in an effective manner, demonstrated through my tenure leading the UTS Rover Team, and my experience working in groups through my university career.

Referees

Contact information available upon request